

Deperdussin 1913 Monocoque Racer

THE DEPERDUSSIN AVIATION COMPANY was set up by Belgian-born businessman Armand Deperdussin in 1910. Deperdussin had made his fortune as a silk importer and knew absolutely nothing about engineering or aeronautics. However, he could spot a good business opportunity, and in the wake of Blériot's Channel crossing and the first Reims air meeting, aviation was a tempting area for entrepreneurs. Deperdussin backed a gifted young engineer, Louis Béchereau, to develop new airplane designs. The Deperdussin 1911 Type C was already an elegant and successful variation on the basic pattern of the famous Blériot XI, but it was in 1912 that Béchereau came up with the revolutionary monocoque design that was to prove the fastest racing model of its time.

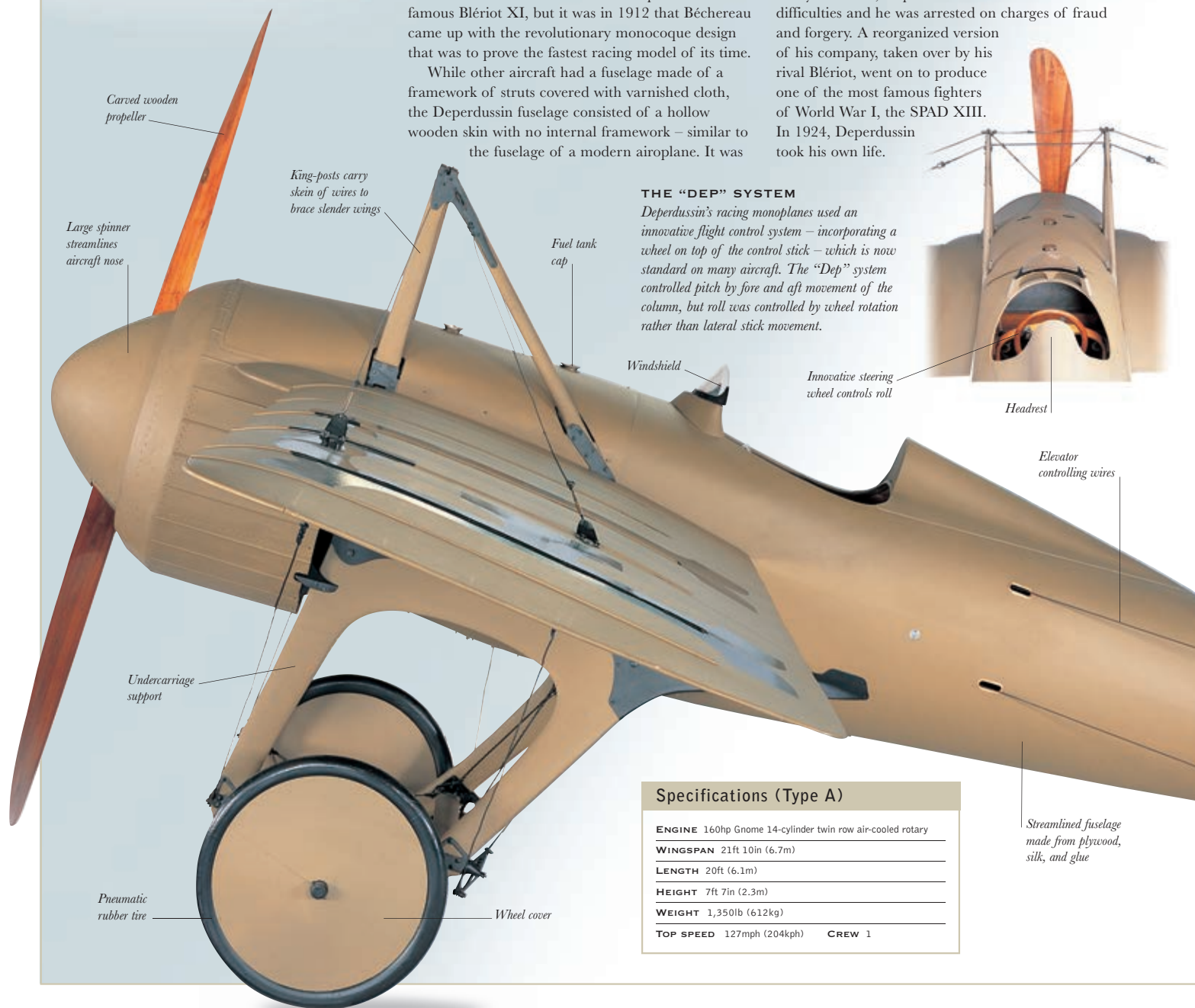
While other aircraft had a fuselage made of a framework of struts covered with varnished cloth, the Deperdussin fuselage consisted of a hollow wooden skin with no internal framework – similar to the fuselage of a modern airplane. It was

also light, streamlined, and, by the standards of the time, carried a powerful engine. In September 1912, Jules Védrières piloted one of the monocoque models to victory in the annual Gordon Bennett race, establishing a new world speed record of 108mph (174kph). The following year, a seaplane version, piloted by Maurice Prévost, won the first Schneider Trophy race at Monte Carlo, and an improved landplane model once again carried off the Gordon Bennett Trophy, as well as establishing a new world speed record of 127mph (204kph).

By that time, Deperdussin had run into financial difficulties and he was arrested on charges of fraud and forgery. A reorganized version of his company, taken over by his rival Blériot, went on to produce one of the most famous fighters of World War I, the SPAD XIII. In 1924, Deperdussin took his own life.

THE "DEP" SYSTEM

Deperdussin's racing monoplanes used an innovative flight control system – incorporating a wheel on top of the control stick – which is now standard on many aircraft. The "Dep" system controlled pitch by fore and aft movement of the column, but roll was controlled by wheel rotation rather than lateral stick movement.



Carved wooden propeller

Large spinner streamlines aircraft nose

King-posts carry skein of wires to brace slender wings

Fuel tank cap

Windshield

Innovative steering wheel controls roll

Headrest

Elevator controlling wires

Undercarriage support

Pneumatic rubber tire

Wheel cover

Streamlined fuselage made from plywood, silk, and glue

Specifications (Type A)

ENGINE	160hp Gnome 14-cylinder twin row air-cooled rotary
WINGSPAN	21ft 10in (6.7m)
LENGTH	20ft (6.1m)
HEIGHT	7ft 7in (2.3m)
WEIGHT	1,350lb (612kg)
TOP SPEED	127mph (204kph)
CREW	1